UNTAN Programming Contest

Problem A

Irene's Apple

Time limit: 1 second
Memory limit: 256 MB



Irene is a beautiful lady who loves gardening and mathematics. She has a vast red apple orchard filled with many apple trees. Her orchard is truly beautiful, with lush greenery and wonderfully fresh, clean air. The apples growing in the orchard are also very ripe and sweet. The orchard is a truly delightful sight, offering a peaceful and refreshing cool atmosphere.

Mia, Aurel, and Lia are Irene's close friends, and they live not far from her orchard. One day, Irene asked them to help her harvest the apples from her beautiful orchard. To make the harvest easier, Irene prepared several large baskets in two colors: green and blue. Each basket was big enough to hold a large number of apples, and only one basket was assigned to each apple tree. The green baskets were meant to hold apples from trees with a prime number of apples, while the blue baskets were used for trees with a non-prime number of apples. After the harvest, Irene wanted to find out how many green baskets contained an even number of apples. She planned to give those baskets to Mia, Aurel, and Lia so they could take them home.

Help Irene by creating a program that determines how many baskets filled with red apples she can give to her friends, Mia, Aurel, and Lia!

Input

– The first line of the input contains an integer N, representing the number of red apple trees in Irene's orchard. The values of N satisfy $1 \leq N \leq 10^5$.

– The second line of the input contains N integers, $A_1, A_2, \ldots, A_N$, where A_i represents the

number of apples harvested from the i-th tree. The values of A_i satisfy $1 \leq A_i \leq 10^9$.

Output

– Print the number of baskets of red apples that Irene will give to Mia, Aurel, and Lia.

Sample Input 1	Sample Output 1
5 50 400 3000 20000 100000	0

Sample Input 2	Sample Output 2
1 1	0